

CH4207

Chemistry of Materials

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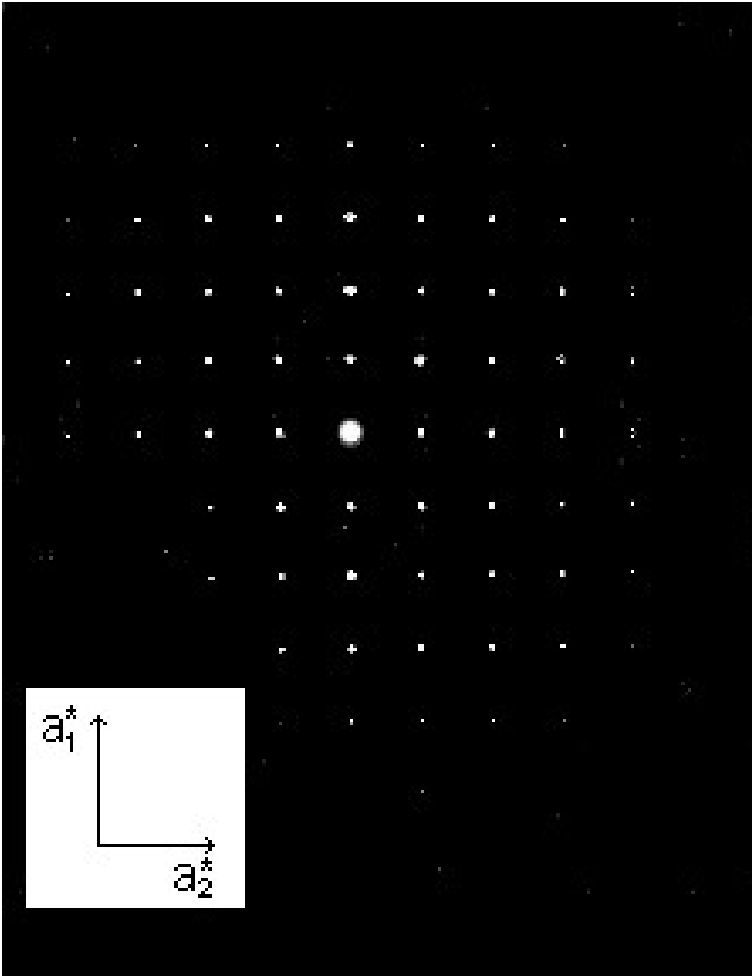
RECIPROCAL LATTICE

Diffraction Pattern

The diffraction pattern is not direct representation of the crystal lattice

Diffraction pattern is a representation of the **reciprocal lattice**

Secondly, Miller indices are derived as reciprocal (inverse) of unit cell intercepts

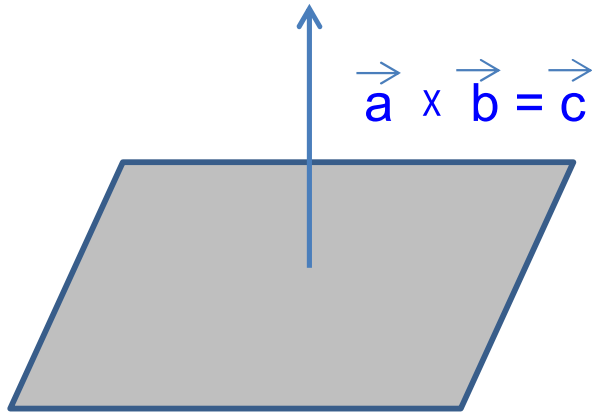


Selected area electron diffraction (SAED) pattern

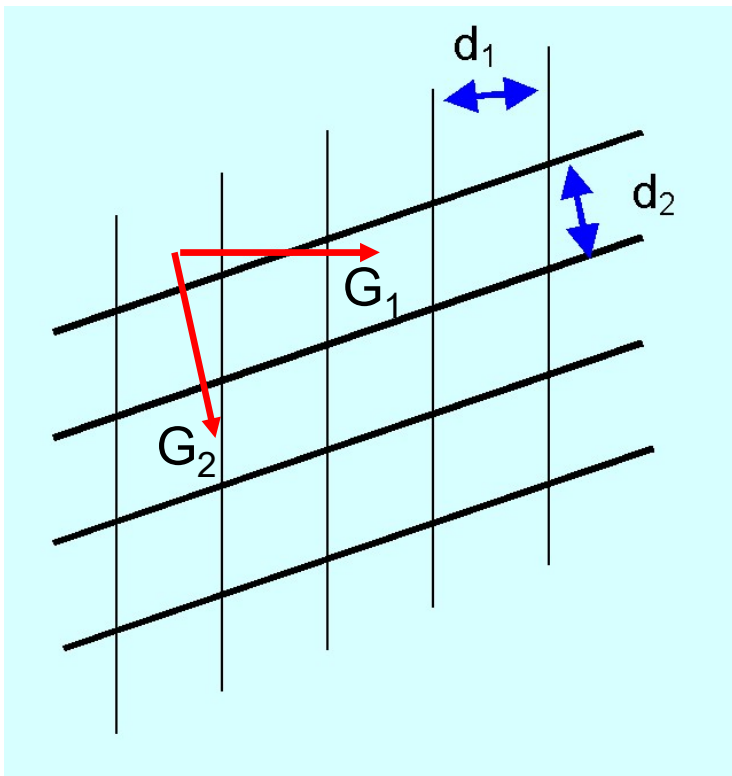
Reciprocal Lattice Vectors

Any set of planes is defined by:

their orientation in the crystal (hkl)
their interplanar spacing (d)



The orientation of a plane is defined by the direction of a normal (vector product)



Let us take two sets of planes.

Draw directions normal.

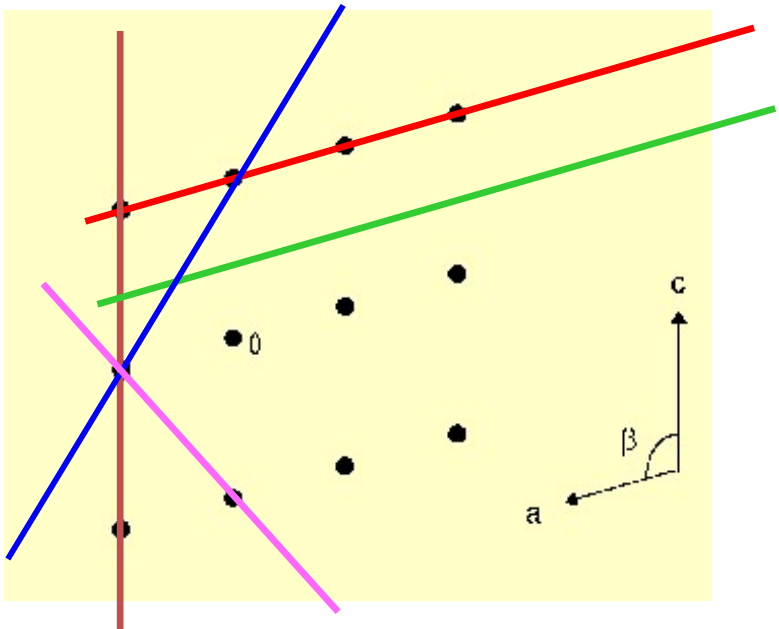
These lines define the orientation but not length

Reciprocal lattice vectors $\vec{G}_1$ and $\vec{G}_2$

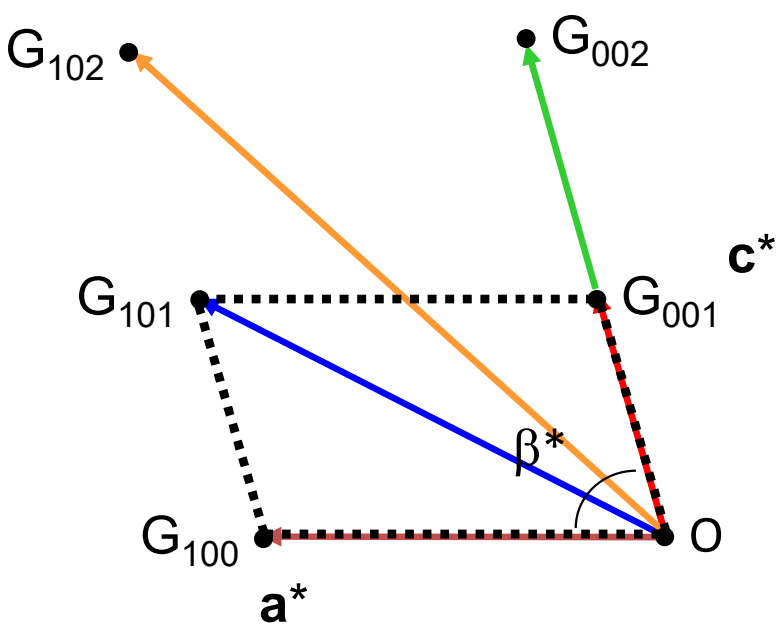
$$\text{Length} = \frac{1}{d}$$

Monoclinic unit cell:

(001) (100) (002) (101) (101̄)



Reciprocal Lattice Vectors

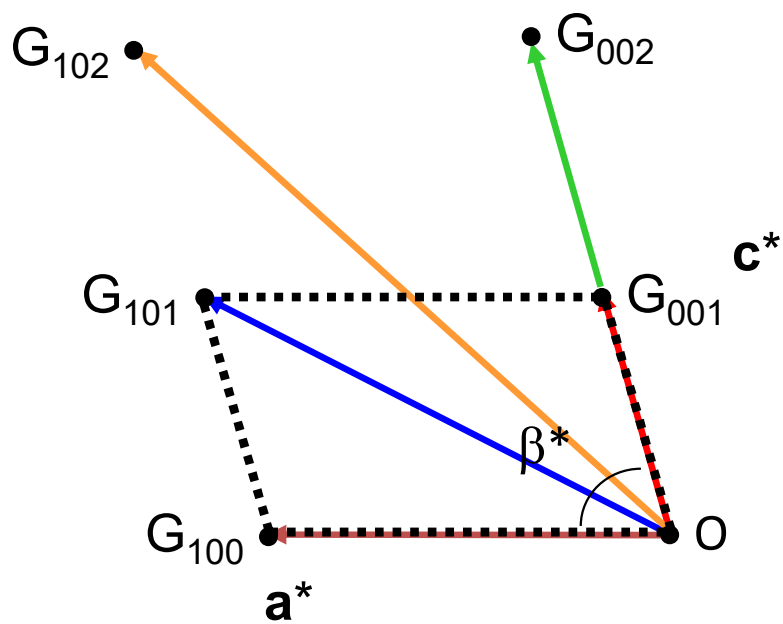


To describe planes, it is important to avoid orthogonal axes.
Use monoclinic unit cell and consider some lattice planes.
These vectors give the outline of the reciprocal unit cell

$\vec{a}^*$ and $\vec{c}^*$ are not parallel to $\vec{a}$ and $\vec{c}$
They are parallel in orthogonal system
 β^* is the complement of β

$$\vec{a}^* = \vec{G}_{100} \quad \text{and} \quad \vec{c}^* = \vec{G}_{001}$$
$$|a^*| = 1/d_{100} \quad \text{and} \quad |c^*| = 1/d_{001}$$

Reciprocal Lattice Vectors



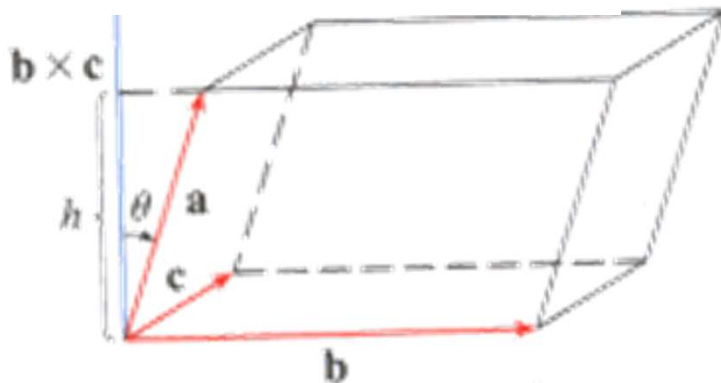
Reciprocal Lattice Vectors

For a lattice denoted by its basis vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, the reciprocal lattice is represented by its basis vectors $\mathbf{a}^*$, $\mathbf{b}^*$ and $\mathbf{c}^*$

$$\mathbf{a}^* = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}$$

$$\mathbf{b}^* = \frac{\mathbf{a} \times \mathbf{c}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}$$

$$\mathbf{c}^* = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}$$



Properties of the Reciprocal Lattice Basis Vectors

1. $\mathbf{a}^* \cdot \mathbf{a} = 1$ (similarly for $\mathbf{b}^*$ and $\mathbf{c}^*$)
2. $\mathbf{a}^*$ is perpendicular to both $\mathbf{b}$ and $\mathbf{c}$, i.e., $\mathbf{a}^* \cdot \mathbf{b} = \mathbf{a}^* \cdot \mathbf{c} = 0$ (similarly for $\mathbf{b}^*$ and $\mathbf{c}^*$) *in orthogonal system*
3. Using an alternative notation where $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_3$ represent the three lattice vectors,

$$\mathbf{a}_i^* \cdot \mathbf{a}_j = \delta_{ij}$$

where δ_{ij} is the Kronecker delta.

Reciprocal lattice vectors have a factor of 2π

Reciprocal Lattice Vectors

While real space vectors are represented by (u, v, w) , reciprocal vectors are given by (h, k, l)

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*$$

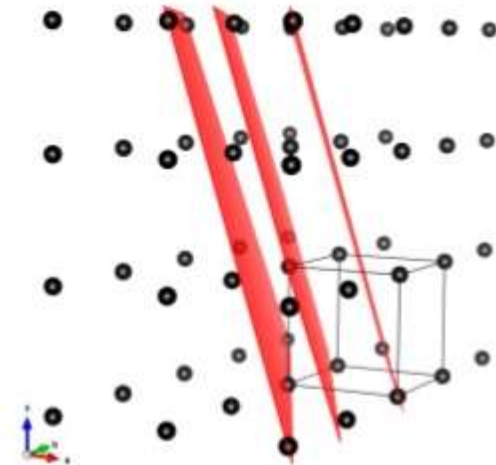
If we consider all real space vectors $\mathbf{r} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$ that are perpendicular to the reciprocal lattice vector:

$$\begin{aligned}\mathbf{r} \cdot \mathbf{G} &= (u\mathbf{a} + v\mathbf{b} + w\mathbf{c})(h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*) \\ &= hu + kv + lw = 0\end{aligned}$$

Equation of lattice planes with Miller indices (h, k, l)

$$\frac{x}{1/h} + \frac{y}{1/k} + \frac{z}{1/l} = N \quad (\text{value determines distance of plane to origin})$$

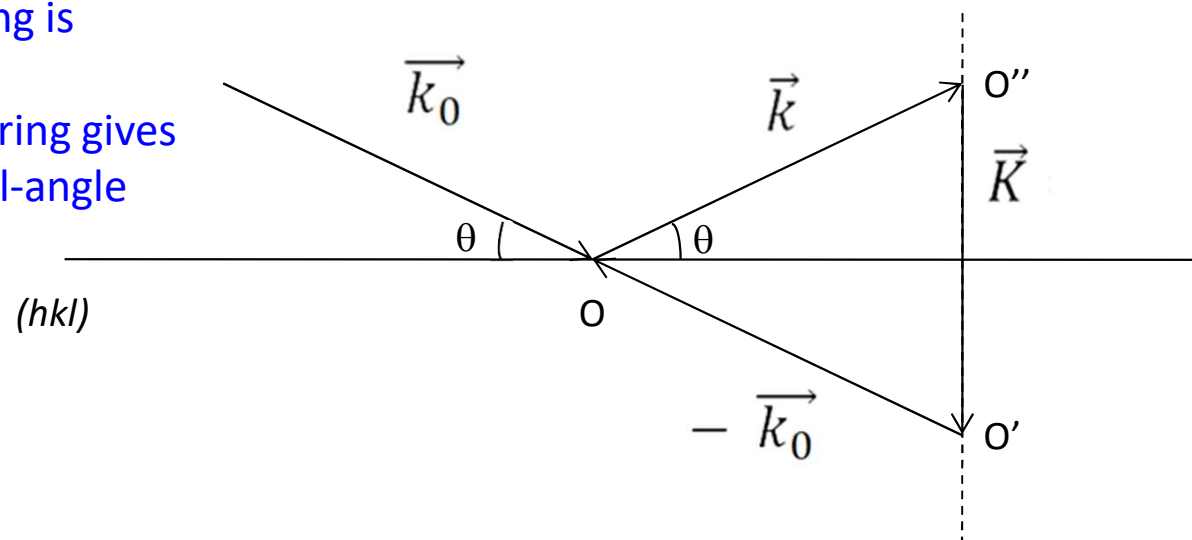
Outcome: The reciprocal lattice vector $\mathbf{G}$ is perpendicular to lattice planes with Miller indices (hkl) and is denoted by $\mathbf{G}_{hkl}$. The reciprocal lattice allows describing plane normals with simple integers.



$\vec{K}$ Vector

Let us define incident X-ray as $\vec{k}_0$ and reflected X-ray as $\vec{k}$, with moduli $1/\lambda$

Elastic scattering is relevant
Inelastic scattering gives negligible small-angle scattering



$$\vec{K} = \vec{k} - \vec{k}_0$$

Difference between scattered wave vector and incident wave vector

Since $\vec{k}$ and $\vec{k}_0$ are of equal length $1/\lambda$, the triangle O, O' and O'' is isosceles.

Angle between $\vec{k}$ and $\vec{k}_0$ is $2\theta_{hkl}$ and the hkl plane bisects it.

Length of $\vec{K}$ (scattering vector):

$$|\vec{K}| = 2|\vec{k}| \sin \theta = \frac{2 \sin \theta}{\lambda}$$

The Laue Condition

$\vec{K}$ is perpendicular to the (hkl) plane, so

$$\vec{K} = \left[\frac{2 \sin \theta_{hkl}}{\lambda} \right] \hat{n}$$

where $\hat{n}$ is a vector of unit length

$\vec{G}$ is perpendicular to the (hkl) plane, so

$$\hat{n} = \frac{G_{hkl}}{|G_{hkl}|}$$

Therefore, $\vec{K} = \frac{2}{\lambda |G_{hkl}|} \sin \theta_{hkl} G_{hkl}$ and since $|G_{hkl}| = \frac{1}{d_{hkl}}$

$$\vec{K} = \frac{2 d_{hkl} \sin \theta_{hkl}}{\lambda} G_{hkl}$$

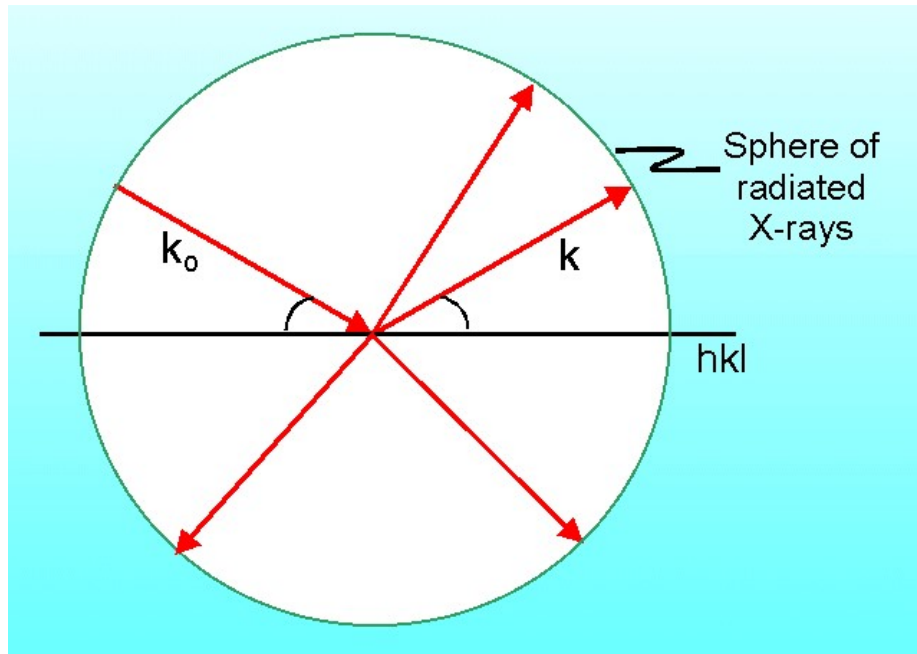
The Laue condition

$$\vec{K} = \vec{G}_{hkl}$$

According to Bragg's law: $2d \sin \theta = \lambda$

Laue condition: significance?

Laue assumed that each set of atoms could radiate the incident radiation in all directions

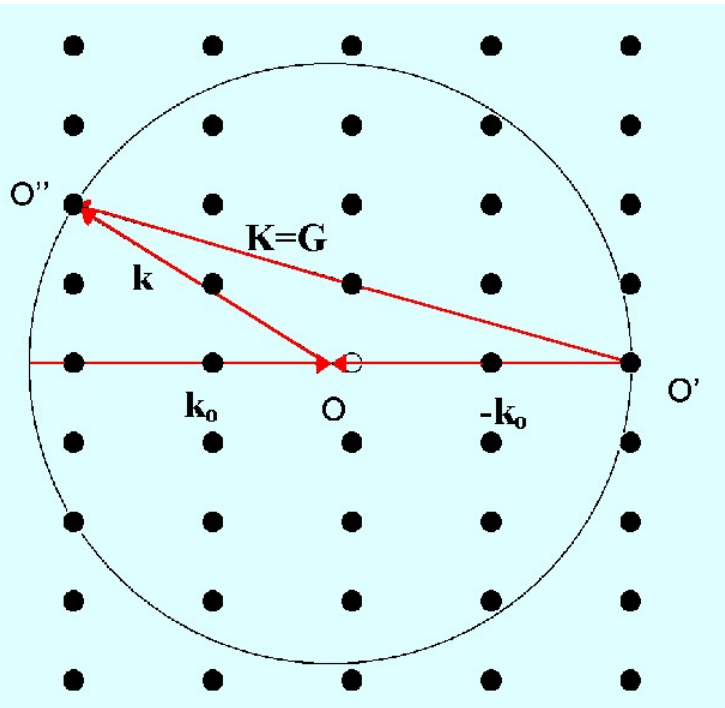


Constructive interference only occurs when the scattering vector, $\vec{K}$, coincides with the reciprocal lattice vector, $\vec{G}$

This leads to **Ewald sphere** construction

Ewald Sphere

We superimpose the imaginary “sphere” of radiated radiation upon the reciprocal lattice

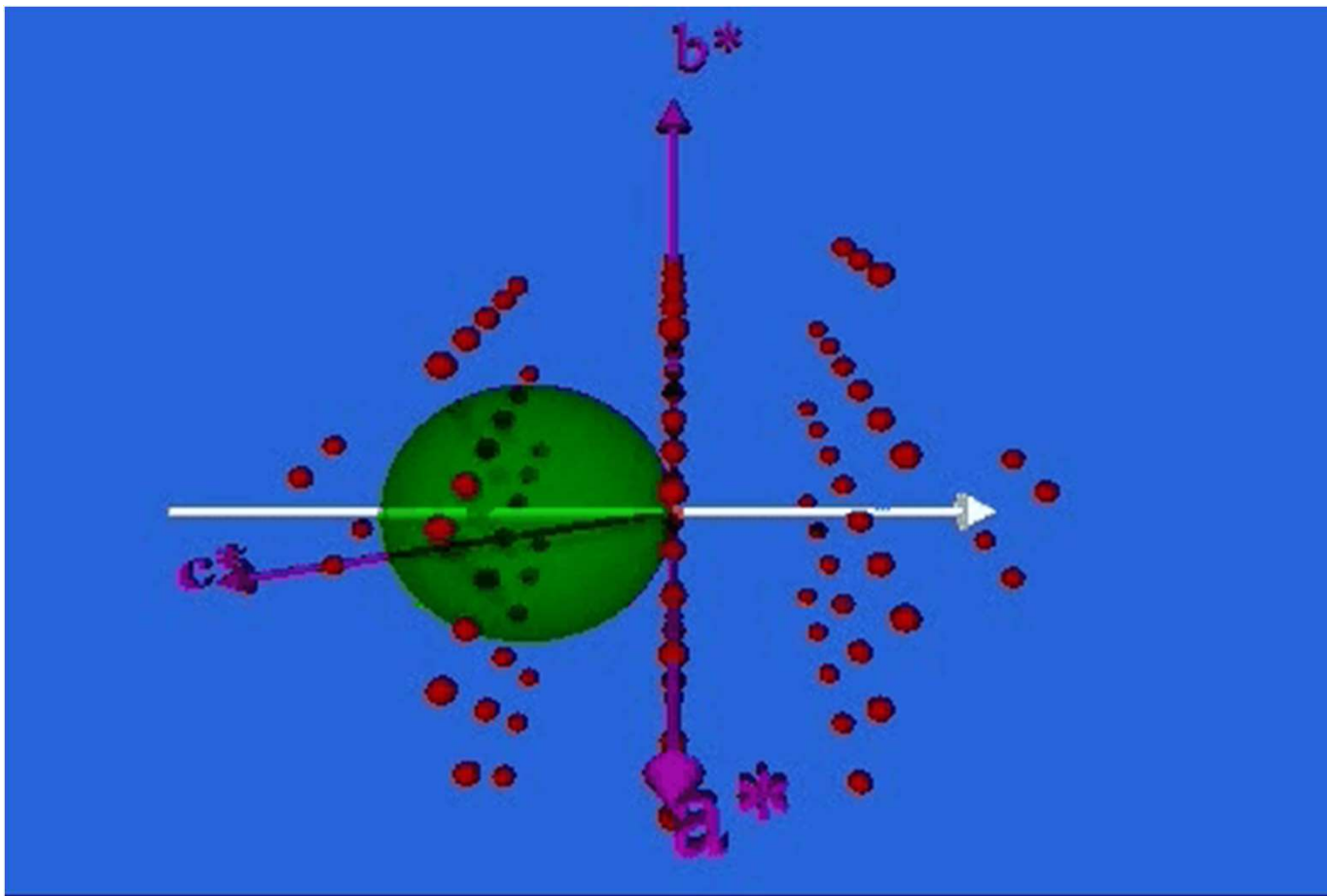


For a fixed direction and wavelength of incident

radiation, we draw $-\vec{k}_0$ ($= \frac{1}{\lambda}$) e.g. along $\vec{a}^*$

Draw sphere of radius $1/\lambda$ centered on end of $\vec{k}_0$
 Reflection is only observed if sphere intersects a
 point i.e. where $\vec{K} = \vec{G}$

Ewald Sphere

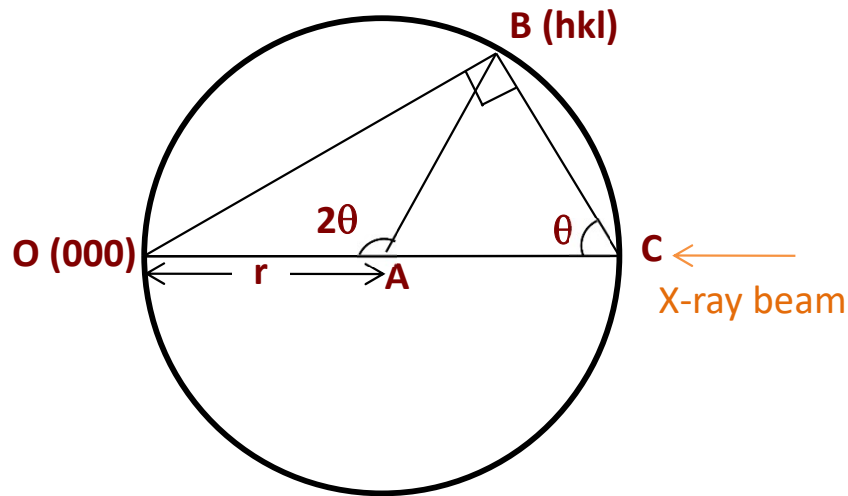


The incident X-ray beam, with wavelength λ , shown as a white line, "creates" an imaginary Ewald's sphere of radius $1/\lambda$ (shown in green). The reciprocal lattice (red points) rotate as the crystal rotates, and every time that a reciprocal point cuts the sphere surface a diffracted beam is produced from the center of the sphere (yellow arrows).

Bragg's law from Reciprocal Lattice

Consider a circle of radius, r , with points B and C lying on the circumference

$$\sin \theta = \frac{OB}{2r}$$



If this geometry is constructed in reciprocal space

The radius can be set to $1/\lambda$, where λ is the experimental wavelength

If O is the (000) reciprocal lattice point, and B is a general point (hkl), then the distance OB is:

$$\frac{1}{d_{hkl}} = S_{hkl}$$

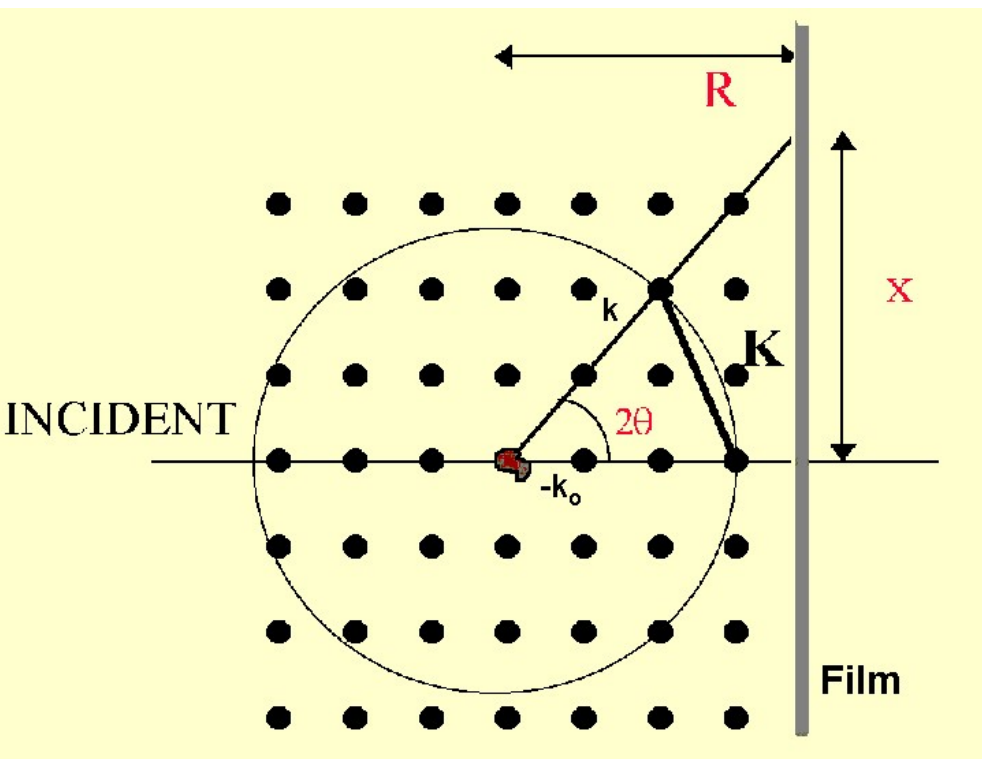
The reciprocal vector between the points, S , increases in magnitude with increasing 2θ

$$\sin \theta = \frac{1}{\frac{d_{hkl}}{2}} \frac{1}{\lambda}$$

$$\lambda = 2d_{hkl} \sin \theta$$

Relate to a Real Diffraction Experiment

The Ewald circle represents in reciprocal space all the possible points where planes (reflections) could satisfy the Bragg equation.



Relating this to a real diffraction experiment with crystal at O

$$\vec{K} = \vec{G}_{hkl}$$

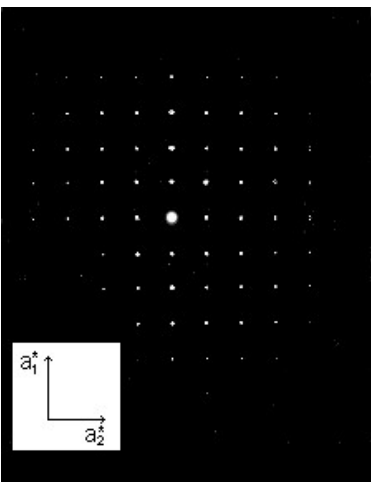
Scattered beam at angle 2θ

Geometry: $\tan 2\theta = \frac{x}{R}$

$$2\theta = \tan^{-1} \frac{x}{R}$$

$$|\vec{K}| = \frac{2 \sin \theta_{hkl}}{\lambda}$$

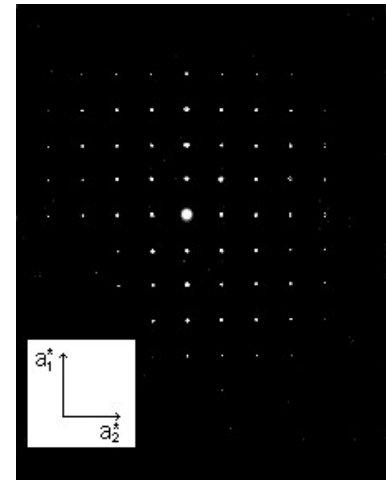
$$|\vec{K}| = \frac{2}{\lambda} \sin \left[\frac{1}{2} \tan^{-1} \frac{x}{R} \right]$$



Relate to a Real Diffraction Experiment

Practicalities:

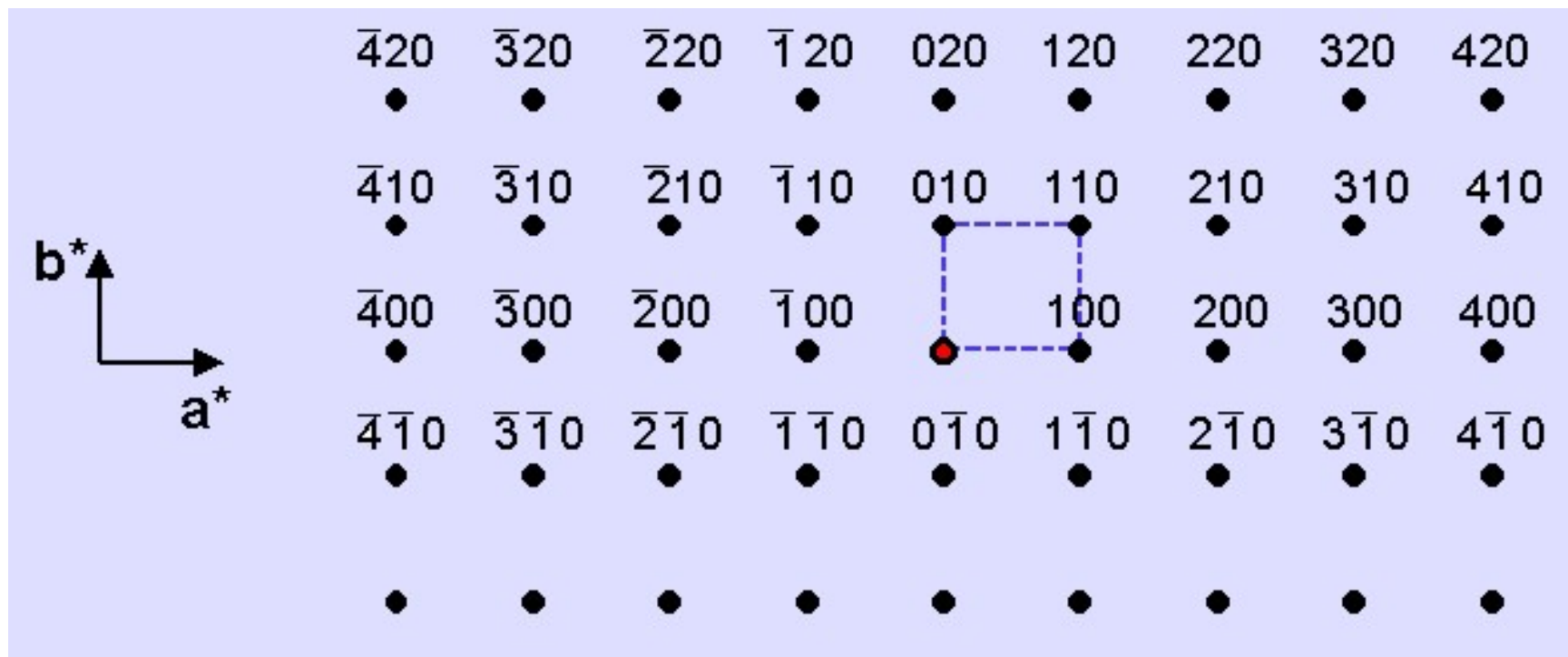
- This allows us to convert distances on the film to lengths of reciprocal lattice vectors
- Indexing the pattern i.e. assigning (hkl) values to each spot allows us to deduce the dimensions of the reciprocal lattice (and hence real lattice)
- In single crystal methods, the crystal is rotated or moved so that each $\vec{G}$ is brought to the surface of the Ewald sphere
- In powder methods, we assume that the random orientation of the crystallites means that $\vec{G}$ takes up all orientations at once



Indexing

Each spot represents a set of planes

Primitive

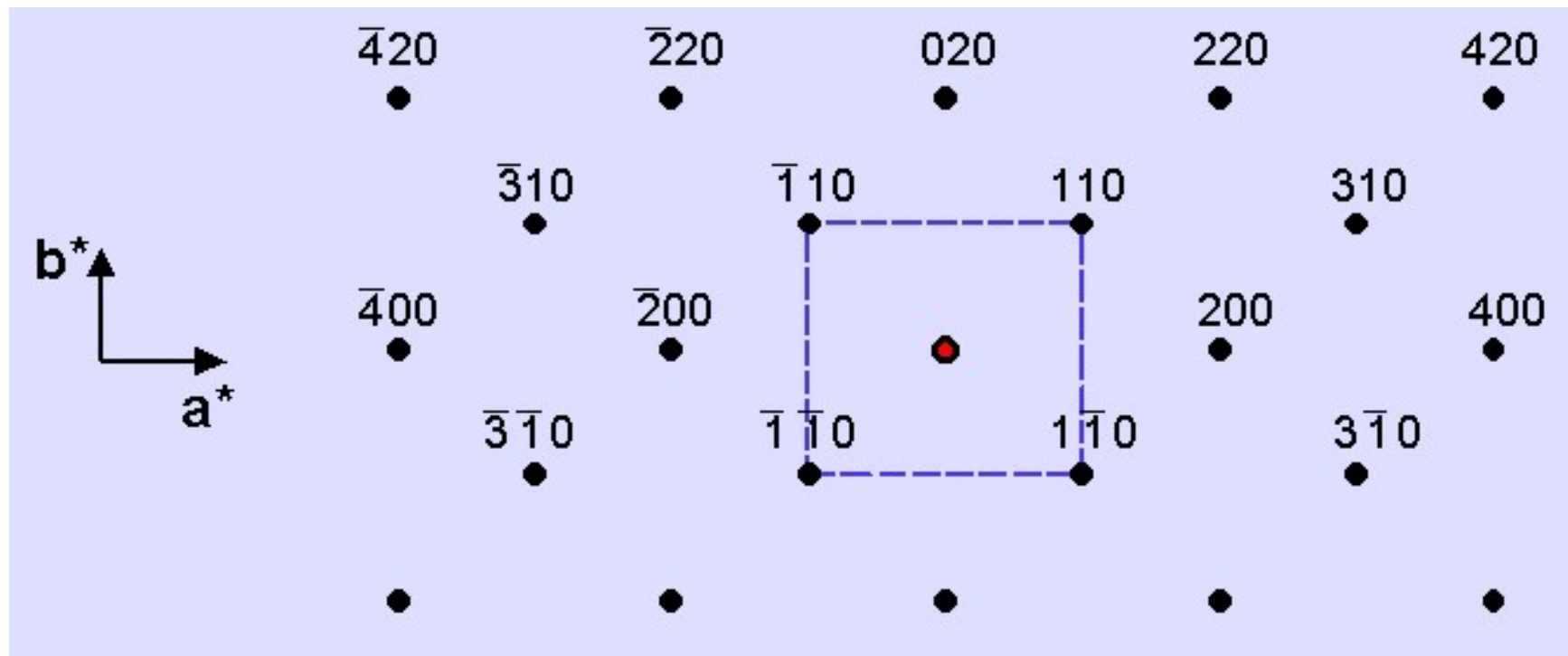


Indexing

Body Centered

$$l = 0$$

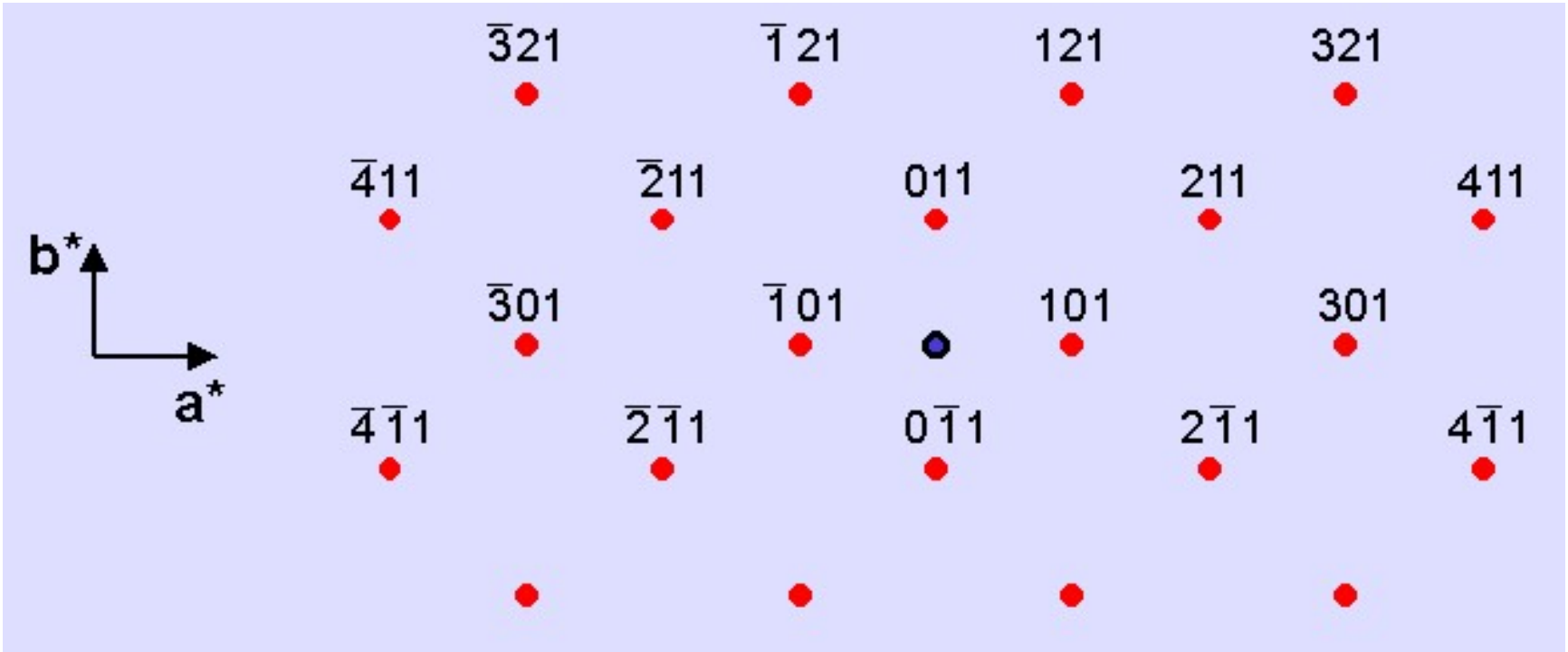
Systematic absences: $h+k+l=2n$



Indexing

Body Centered

Systematic absences: $h+k+l=2n$ l = 1

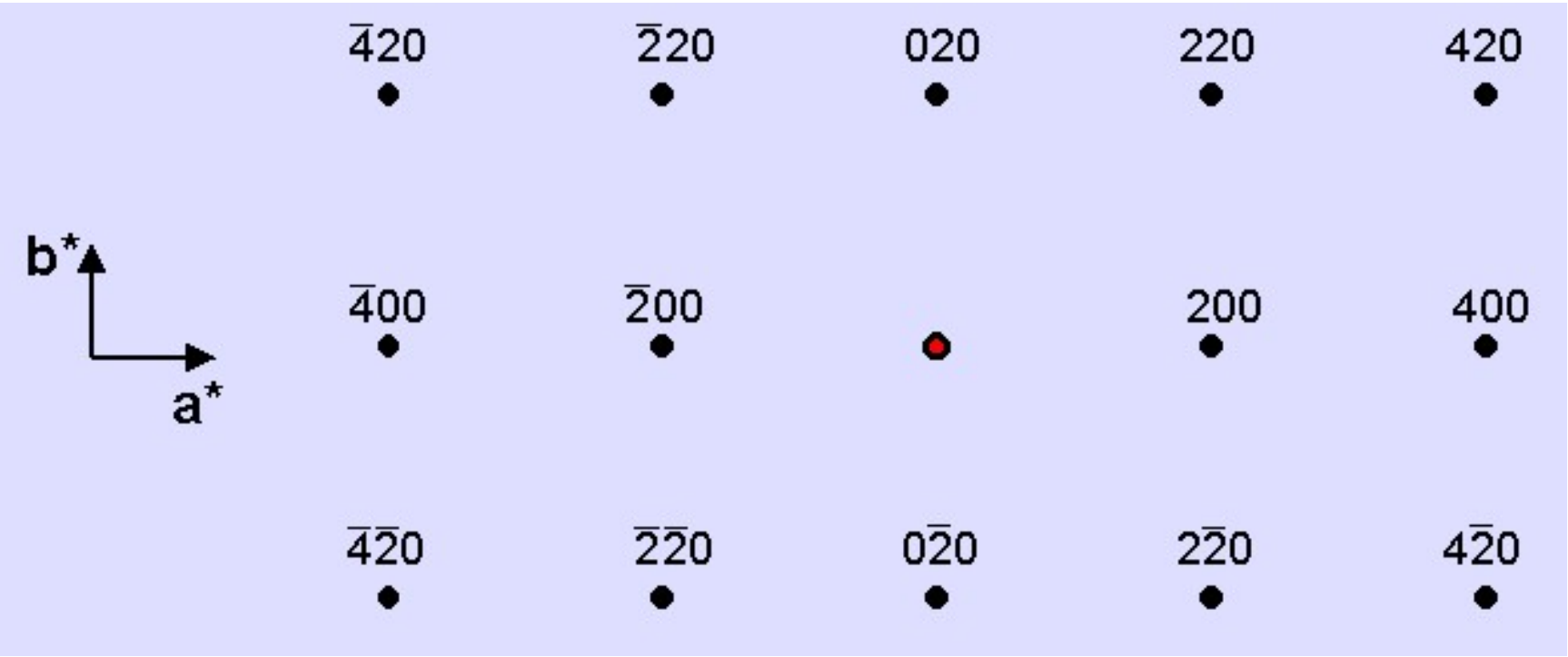


Indexing

Face Centered

$$I = 0$$

Systematic absences: h,k,l all odd or all even



Indexing

Face Centered

$l = 1$

Systematic absences: h,k,l all odd or all even

